

Answers: Mean Class Size

Suppose a college is offering five sections of an introductory statistics course. The class sizes for the sections are:

Section	301	302	303	304	305
Class size	10	4	5	3	6

a) Find the mean class size per section. What is the observational unit? What is the variable?

The observational unit is a **section** (there are 5). The variable is the **class size** of the section.

$$\bar{x} = \frac{10 + 4 + 5 + 3 + 6}{5} = \frac{28}{5} = 5.6 \text{ students}$$

b) Find the mean class size per student, using the class sizes the students report. What is the observational unit? What is the variable?

The observational unit is a **student** (there are 28). The variable is the **class size of the student's section**, as the student reports it. Each student reports the class size of their own section:

Section	Students reporting	Value each reports	Total
301	10	10	100
302	4	4	16
303	5	5	25
304	3	3	9
305	6	6	36
All	28		186

$$\bar{x} = \frac{186}{28} \approx 6.64 \text{ students}$$

The mean is bigger from the students' point of view. More students are in the bigger sections (10 of the 28 are in section 301), so a typical student is in a bigger section than a typical section has. Which mean you get depends on what the observational units are.